Lagrangian Models for Ocean Transport
15-17 June, 2026

Activity 1: Create a proper environment for MOHID

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MOHID-Lagrangian environment

- MOHID-Lagrangian framework is composed of **two parts**:
  - The **numerical core** , written in Fortran 
  - The **pre- and post-processor**, implemented as Python scripts

Pre- and Post-Processor:

What they are?

- **Python workflow tools**
  - Parse standard HDF5 & NetCDF-CF files

What are they built?

- Python 3.13 or higher
  - Use several libraries as pandas, matplotlib, shapefile, etc

What do we use?

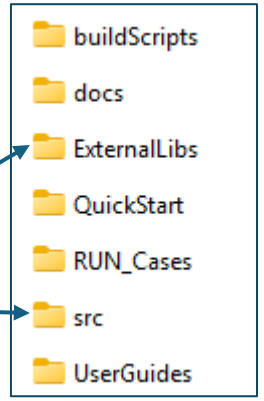
- **Anaconda:**
  - **Miniconda**, a lightweight minimal alternative

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Numerical Core:

What is it?

- It processes large Lagrangian tracer ensembles, executing critical algorithms rapidly.
- It is broken up in:
  - **MOHID Lagrangian** sources
  - **External Libraries Sources**, including HDF5 and NetCDF.

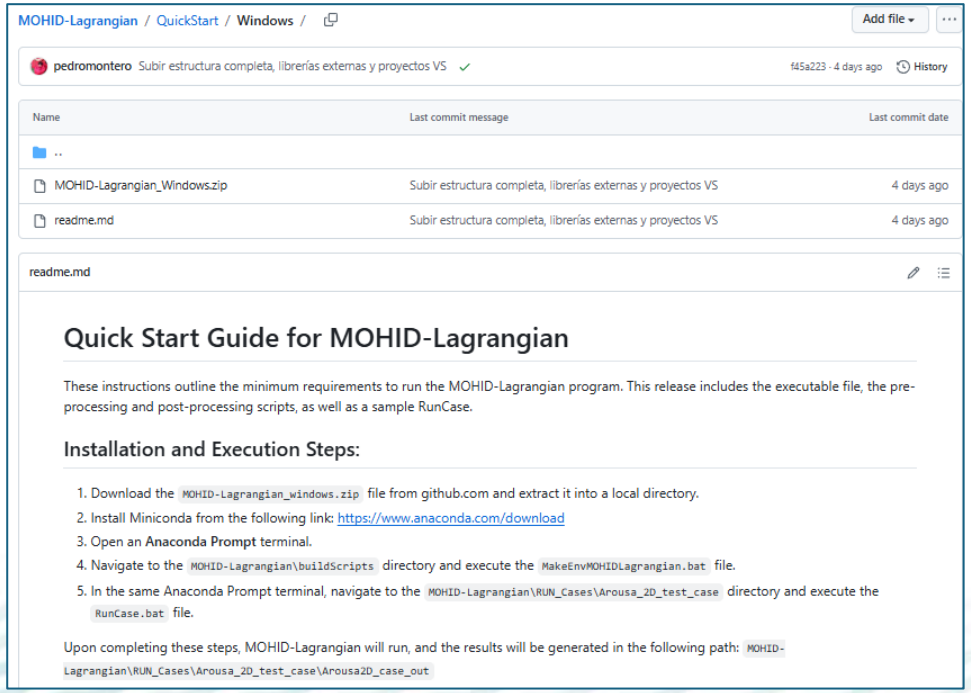


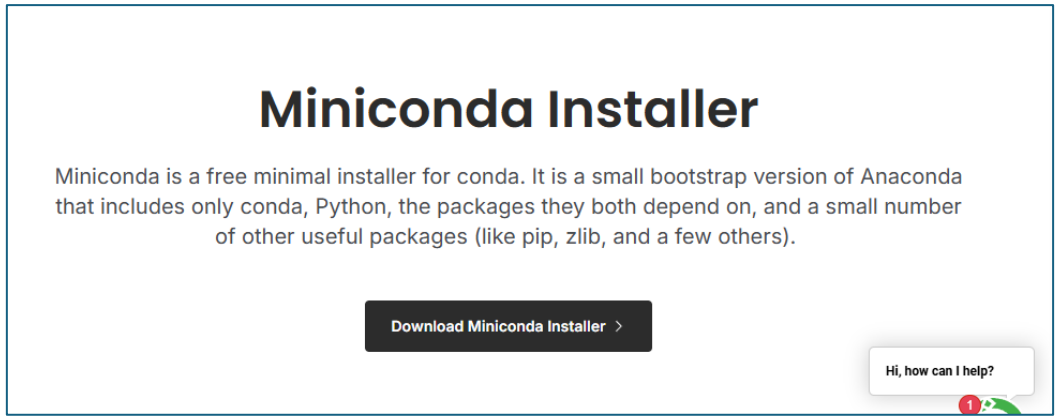
What are they built?

- **Fortran version:** Intel Fortran
- Nowadays, since HDF5 must be HDF5 1.8.17, it **must be compiled** with **Intel Fortran ifort 2013**. Incompatible with new version ifx 2026.
- **Future:** Update HDF5 libraries.

Windows Setup (I):

- Visit Github repository: https://github.com/Mohid-Water-Modelling-System/MOHID-Lagrangian
- Go to QuickStart-> Windows
- Download MOHID-Lagrangian_windows.zip and uncompress it
- Install Miniconda: https://www.anaconda.com/download

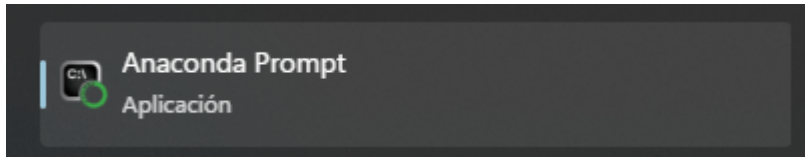




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Windows

Windows Setup (II):



- Open an **Anaconda Prompt** terminal.
- Navigate to the MOHID-Lagrangian\buildScripts directory and execute the **MakeEnvMOHIDLagrangian.bat** file.

```
 Directorio de C:\Users\UsrXModel1\Documents\02_TRABALLO\03_MODELOS\02_MOHID_LAGRANGIAN\
\buildScripts

04/06/2026  09:27    <DIR>          .
04/06/2026  11:21    <DIR>          ..
04/06/2026  09:27             1.081 1-install_ExternalLibs.sh
04/06/2026  09:27             2.704 2-compile_Lagrangian.sh
04/06/2026  09:27               525 3-MakeEnvMOHIDLagrangian_noconda.sh
04/06/2026  09:27             2.963 environmentLinux.yml
04/06/2026  09:27             2.542 environmentWindows.yml
04/06/2026  09:27               579 MakeEnvMOHIDLagrangian.bat
04/06/2026  09:27               579 MakeEnvMOHIDLagrangian.sh
04/06/2026  09:27               718 MakeEnvMOHIDLagrangian_anaconda.sh
04/06/2026  09:27               473 MakeEnvMOHIDLagrangian_noconda.bat
04/06/2026  09:27               525 MakeEnvMOHIDLagrangian_noconda.sh
04/06/2026  09:27               949 MakeMOHIDLagrangian.sh
04/06/2026  09:27               817 README.txt
04/06/2026  09:27                74 requirements.txt
04/06/2026  09:27               844 RunCmakeMOHIDLagrangian.bat
              14 archivos         15.373 bytes
               2 dirs  74.947.280.896 bytes libres
```

- In the same Anaconda Prompt terminal, navigate to the MOHID-Lagrangian**RUN_Cases\Arousa_2D_test_case** directory and execute the **RunCase.bat** file.
- This quickstart has only a RUN_Cases, to download the other ones, revisit the github page.

Linux Setup(I):

Unlike Windows, the Linux approach involves downloading the full project, compiling libraries/models, and installing a Conda environment for pre/post-processing

OS Compatibility

- **Native Support:** Ubuntu/Debian (via apt) and Rocky Linux/RHEL (via dnf).
- **Universal Fallback:** Works on any other Linux distribution (Arch, openSUSE, etc.) if `git`, `python3`, `gcc`, and `vim` are pre-installed.
- **Isolated Stack**: Spack and Miniconda handle all model dependencies independently.

Setup

- Visit Github repository: https://github.com/Mohid-Water-Modelling-System/MOHID-Lagrangian
- Go to QuickStart-> Linux
- Download MOHIDLagrangian_install.sh

Linux Setup(II):

Edit this file and change the options:

```
#!/bin/bash
#================================================================================
#     MULTI-OS INSTALLER FOR MOHID-LAGRANGIAN (UBUNTU & ROCKY LINUX)
#================================================================================
# This script installs MOHID-Lagrangian and all its dependencies.
# Compatible with:
#   - Debian/Ubuntu based systems (apt)
#   - Rocky Linux/RHEL/CentOS based systems (dnf/yum)
#================================================================================

# --- Error handling: the script will stop if a command fails ---
set -e

#================================================================================
#     CONFIGURATION (MODIFY IF NECESSARY)
#================================================================================

# 0. PERMISSIONS ROOT SETTINGS
#    Defines the folder name inside your home directory.
#    Set to true to use sudo for system dependencies.
#    If false, the script will report missing packages to be installed by an admin.
#    - Default: "false"
USE_SUDO=false

# 1. INSTALLATION DIRECTORY NAME
#    Defines the user has sudo privileges.
#    - Default: "False" -> installs to $HOME/lagrangian
INSTALL_FOLDER_NAME="lagrangian"

# 2. GITHUB BRANCH
#    Define the branch you want to clone.
#    - Default: "master"
TARGET_BRANCH="master"

# 3. COMPILE EXTERNAL LIBRARIES?
#    Set to true to compile ExternalLibs (required for first installation).
#    Set to false to skip this step (if they are already compiled).
COMPILE_LIBS=true

# 4. BUILD TYPE
#    Options: "RELEASE" or "DEBUG"
BUILD_TYPE="RELEASE"

# 5. TOOL VERSIONS (Spack)
INTEL_COMPILER_VERSION="intel-oneapi-compilers@2023.2.1"
MINICONDA_VERSION="miniconda3@24.3.0"

# 6. PROJECT FOLDER NAME (Internal repo folder name)
PROJECT_DIR_NAME="MOHID-Lagrangian"
```

Linux Setup(III):

Options:

Variable	Default	Description
USE_SUDO	false	Set to true to allow automatic installation of system packages. If false, it will display the required commands for manual installation by an admin, of git, gcc, vim,python3 and which
INSTALL_FOLDER_NAME	"lagrangian"	The directory name inside your $HOME folder where everything will be installed.
TARGET_BRANCH	"master"	The specific GitHub branch of the repository you want to clone.
COMPILE_LIBS	true	Whether to compile the mandatory external libraries. Must be true for the first installation.
BUILD_TYPE	"RELEASE"	Compilation mode. Use "RELEASE" for maximum performance or "DEBUG" for development and troubleshooting.
INTEL_VERSION	intel-oneapi-compilers	The version of the Intel compiler to be managed and installed via Spack.
MINICONDA_VERSION	miniconda3	The Miniconda version to be installed via Spack for Python environment management.

Linux Setup(IV):

How to Execute:

Grant Execution Permissions: `chmod +x MOHIDLagrangian_install.sh`
Run the Script: `./MOHIDLagrangian_install.sh`
Finalize: Run `source ~/.bashrc` after completion to update your current terminal session paths.

How to run:

```
conda activate MOHID-Lagrangian
cd /RUN_Cases/Tagus3D_case
./RunCase.sh
```

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How to compile it in Windows:

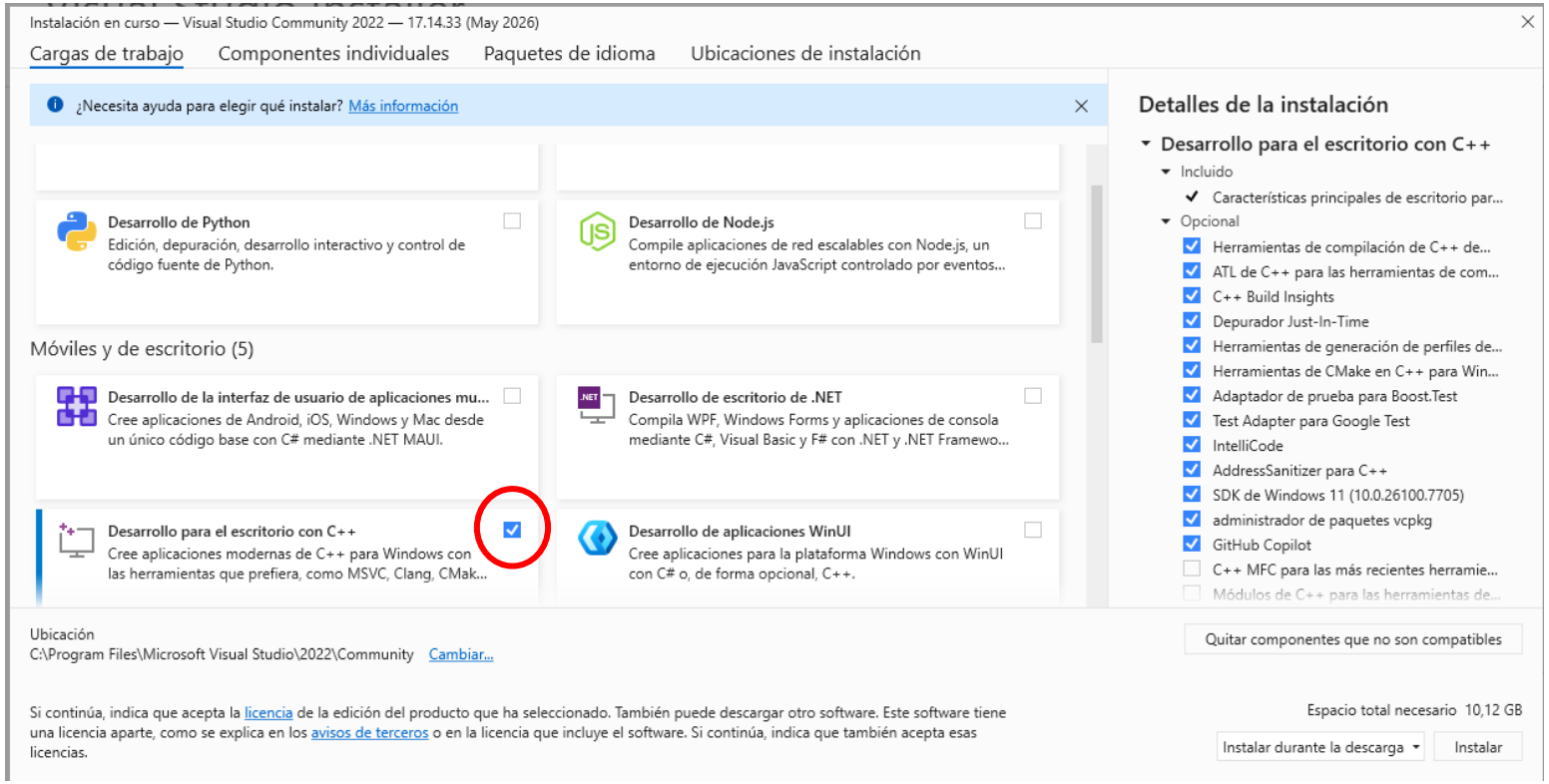
- **Compiler Requirement**: Must compile with `Intel ifort`.
- **Issue**: Intel `ifx` causes critical failures when handling `OpenMPI`.
- **Prerequisites:** Requires Visual Studio 2020 and Intel `ifort 2023` sources (do not use Visual Studio 2026 or `ifx`).

Executables to setup:

Visual-Studio-2022-Windows-17-3-5-es.exe
Instalar w_BaseKit_p_2023.2.0.49396_offline.exe
Instalar w_HPCKit_p_2023.2.0.49441_offline.exe

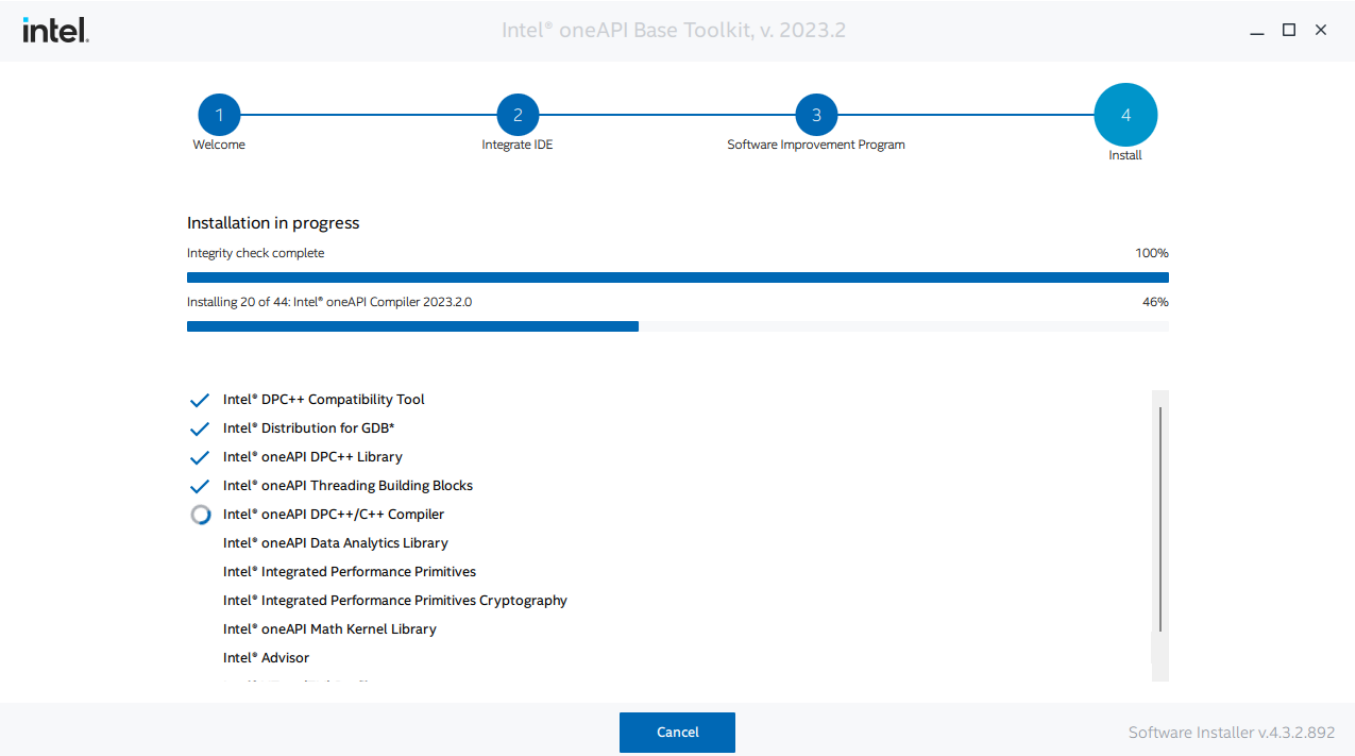
How to compile it in Windows (prerequisites):

- **Visual-Studio-2022-Windows-17-3-5-es.exe**



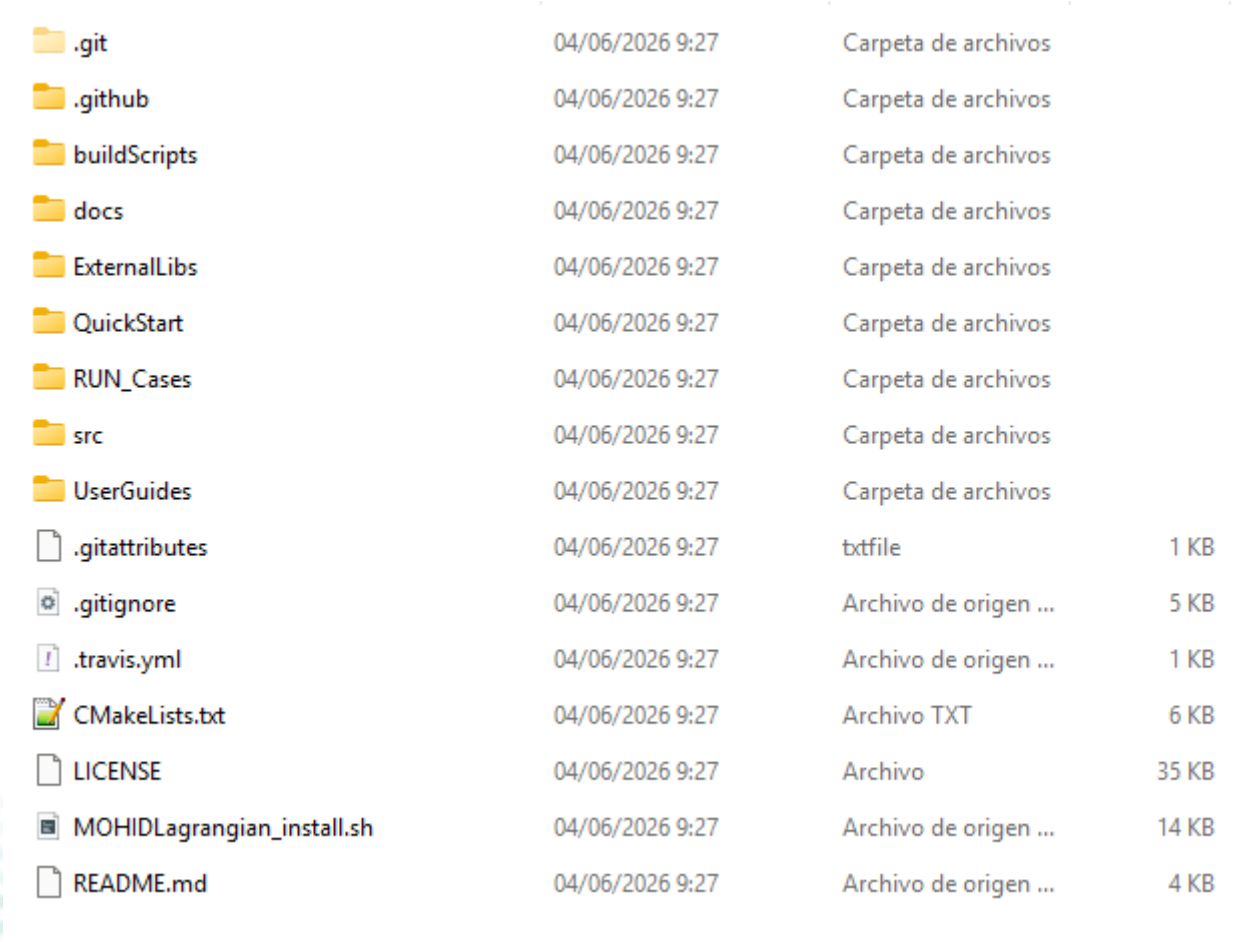
How to compile it in Windows (prerequisites):

- w_BaseKit_p_2023.2.0.49396_offline.exe
- w_HPCKit_p_2023.2.0.49441_offline.exe



How to compile it in Windows:

- Visit Github repository: https://github.com/Mohid-Water-Modelling-System/MOHID-Lagrangian
- Download or git clone code in a folder:

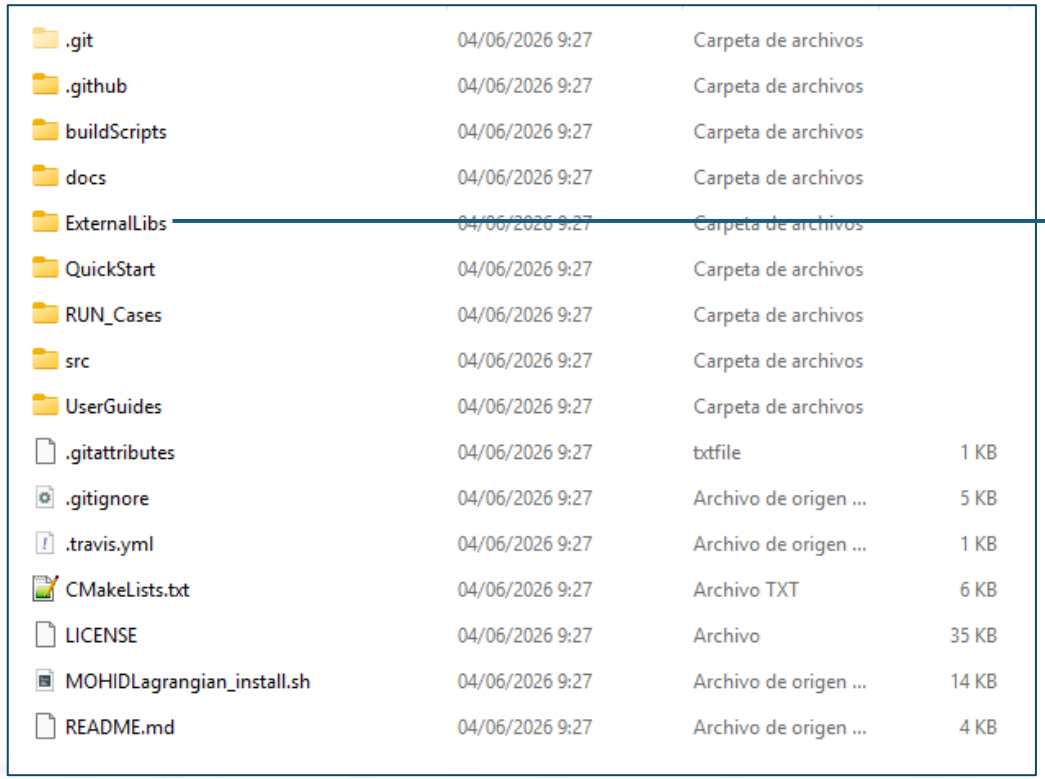


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.github	04/06/2026 9:27	Carpeta de archivos	
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CMakeLists.txt	04/06/2026 9:27	Archivo TXT	6 KB
LICENSE	04/06/2026 9:27	Archivo	35 KB
MOHIDLagrangian_install.sh	04/06/2026 9:27	Archivo de origen ...	14 KB
README.md	04/06/2026 9:27	Archivo de origen ...	4 KB

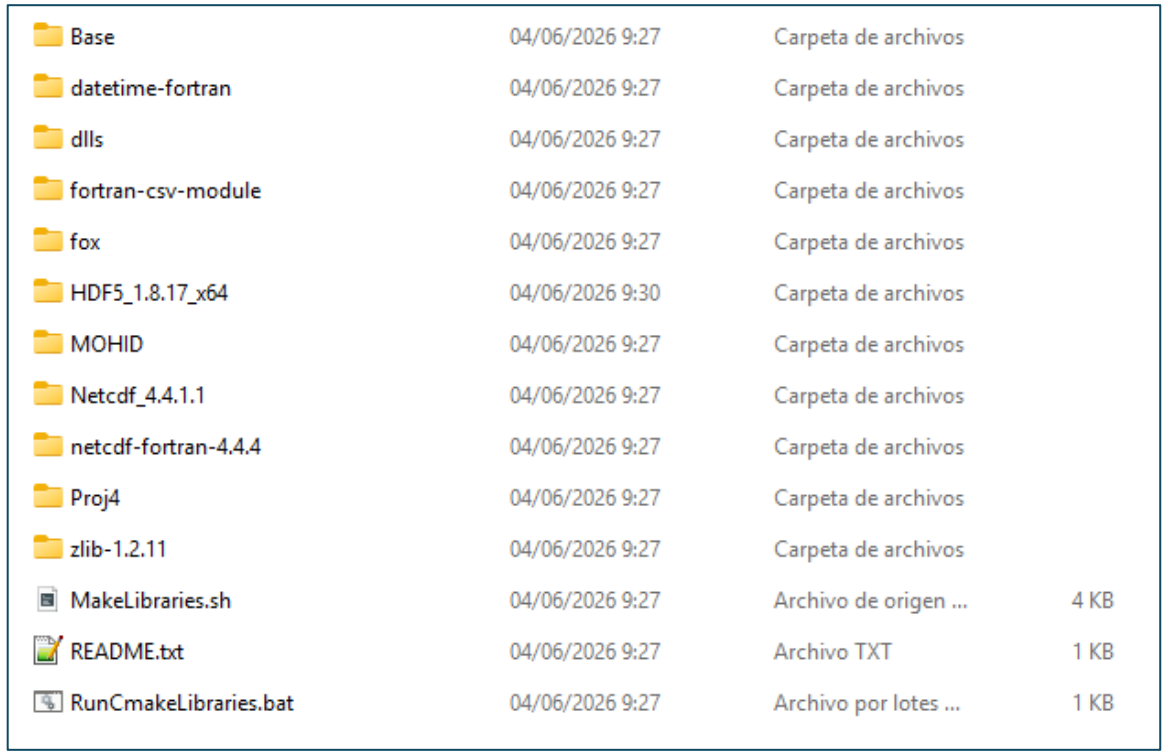
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Compiling external libs:

- Open ExternalLibs folder



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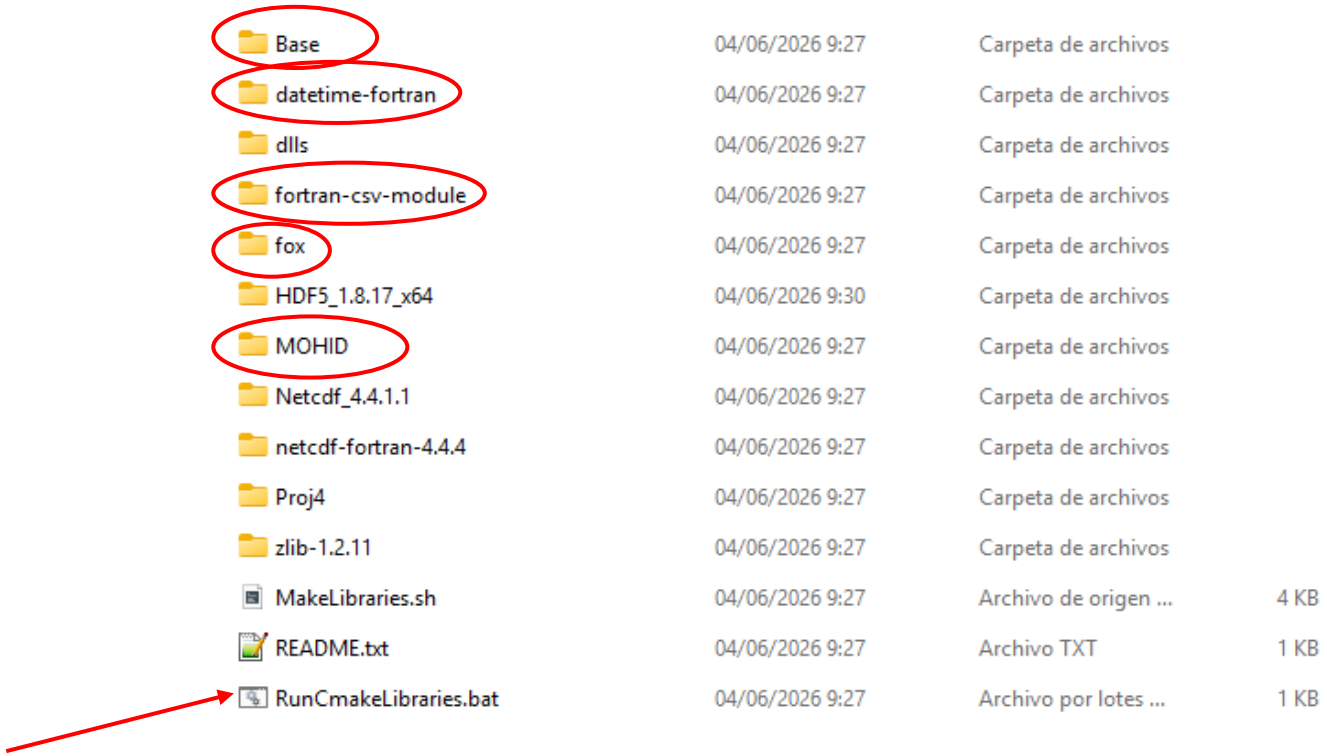


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datetime-fortran	04/06/2026 9:27	Carpeta de archivos	
dlls	04/06/2026 9:27	Carpeta de archivos	
fortran-csv-module	04/06/2026 9:27	Carpeta de archivos	
fox	04/06/2026 9:27	Carpeta de archivos	
HDF5_1.8.17_x64	04/06/2026 9:30	Carpeta de archivos	
MOHID	04/06/2026 9:27	Carpeta de archivos	
Netcdf_4.4.1.1	04/06/2026 9:27	Carpeta de archivos	
netcdf-fortran-4.4.4	04/06/2026 9:27	Carpeta de archivos	
Proj4	04/06/2026 9:27	Carpeta de archivos	
zlib-1.2.11	04/06/2026 9:27	Carpeta de archivos	
MakeLibraries.sh	04/06/2026 9:27	Archivo de origen ...	4 KB
README.txt	04/06/2026 9:27	Archivo TXT	1 KB
RunCmakeLibraries.bat	04/06/2026 9:27	Archivo por lotes ...	1 KB

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Compiling external libs:

- **Target Libraries:** Compile only Base, datetime-fortran, fortran-csv-module, fox, and MOHID (others are already compiled).

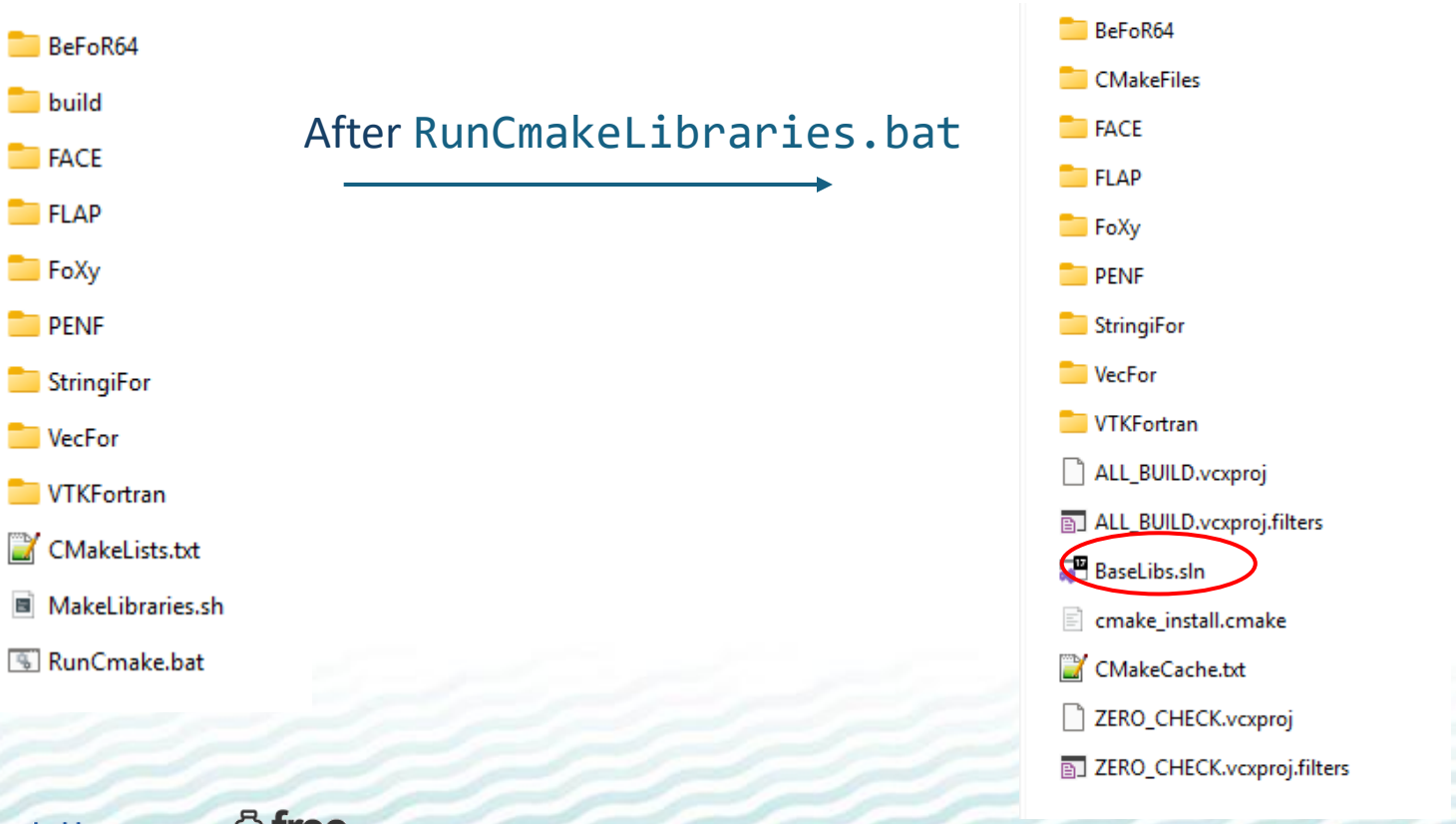


- Step 1 - Project Generation: Run `RunCmakeLibraries.bat` from the root directory to automatically generate `.sln` projects across all 5 library folders.

Compiling external libs:

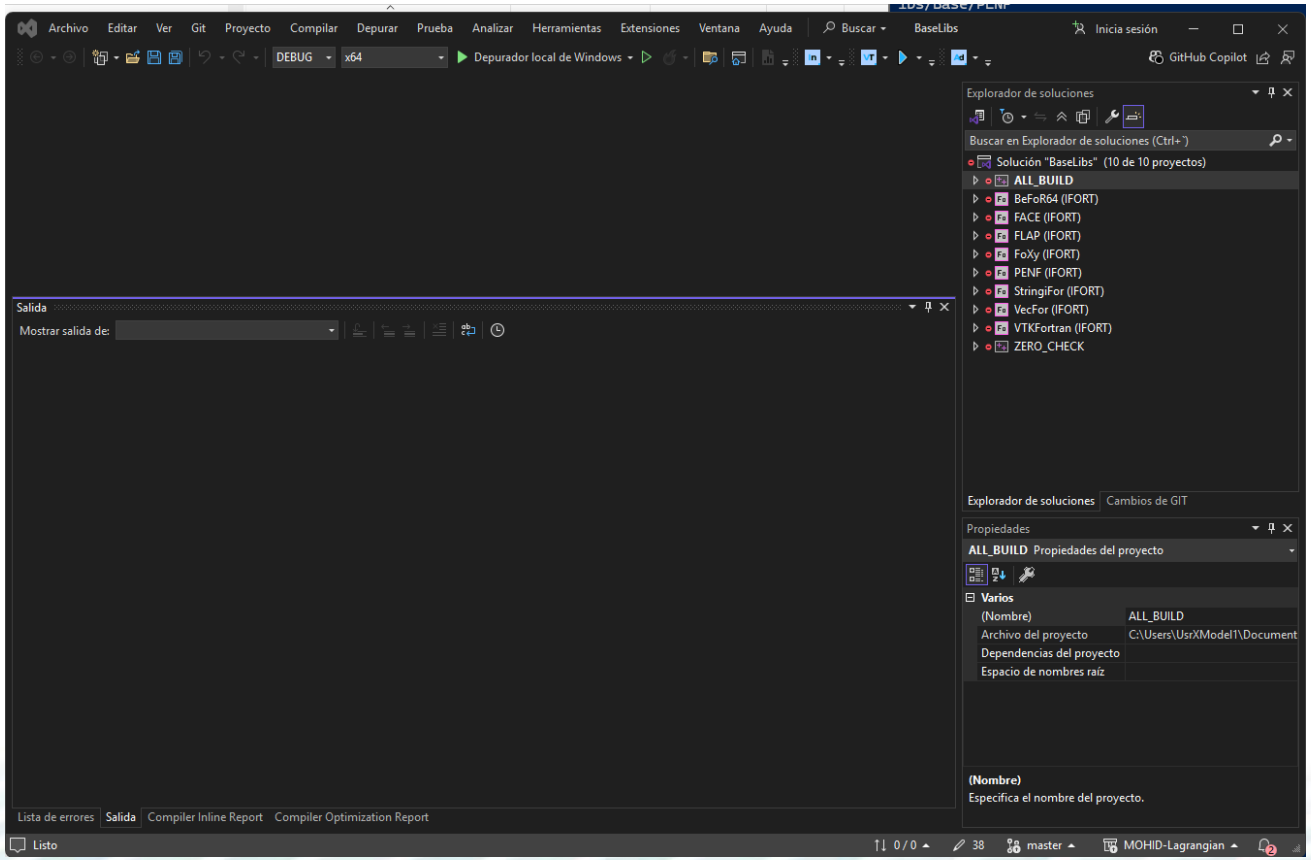
- Step 1 - Project Generation: Run `RunCmakeLibraries.bat` from the root directory to automatically generate .sln projects across all 5 library folders.

- Ex: `Base` folder



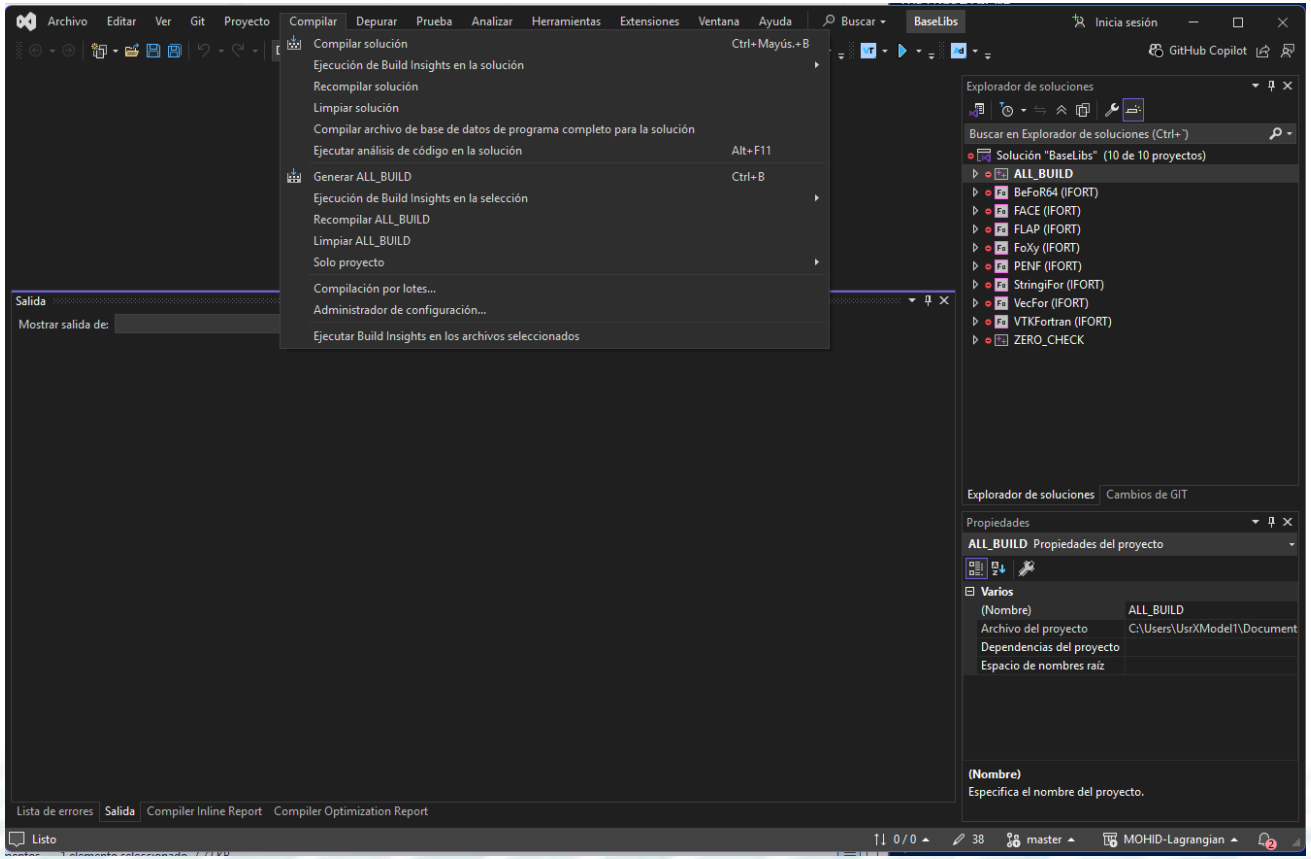
Compiling external libs:

•Step 2 - Compilation: For each library, navigate to its new build/ folder, open the .sln file in Visual Studio 2022, and build in both DEBUG and RELEASE modes.



Compiling external libs:

•Step 2 - Compilation: For each library, navigate to its new build/ folder, open the .sln file in Visual Studio 2022, and build in both DEBUG and RELEASE modes.

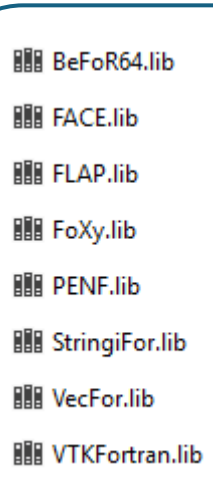


Windows

Compiling external libs:

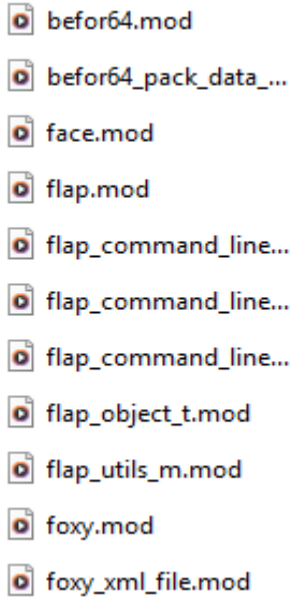
- **After compiling libraries:** There are two new folders in a build folder for each library:

lib



- **What they are:** Compiled static libraries containing the ready-to-use binary code.
- **What they are for**: They merge into the final executable to make it completely independent.

modules

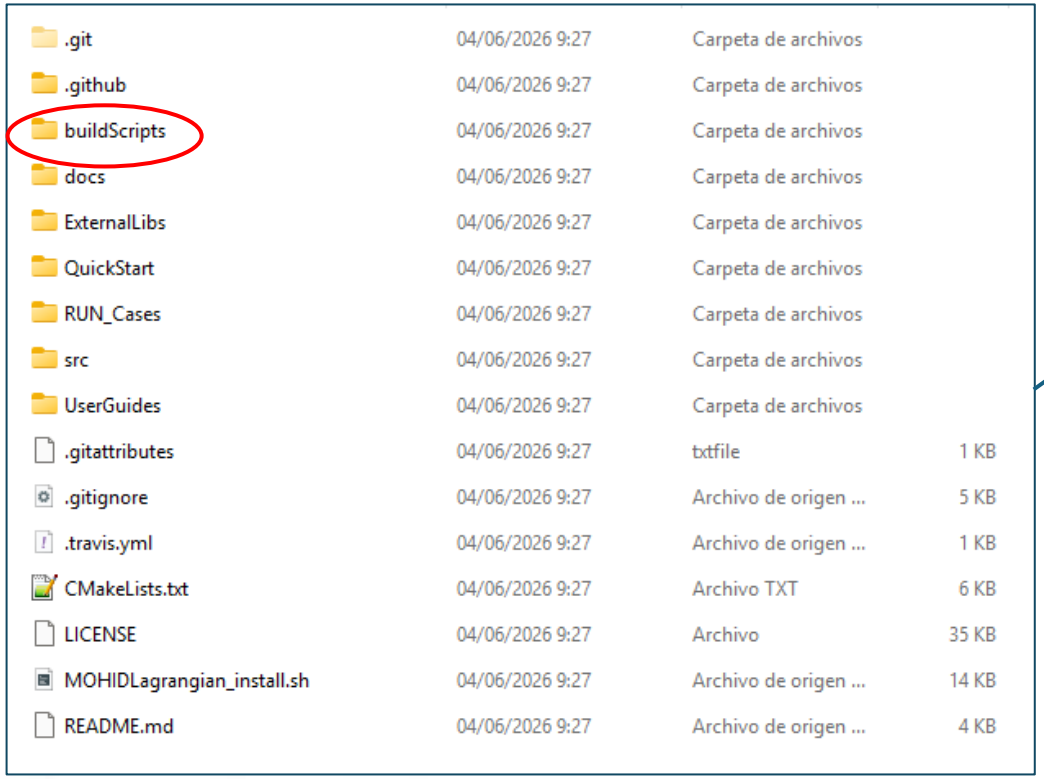


- **What they are:** Compiled Fortran modules that act like C/C++ header files.
- **What they are for**: They let the compiler verify function arguments and types during a USE statement.

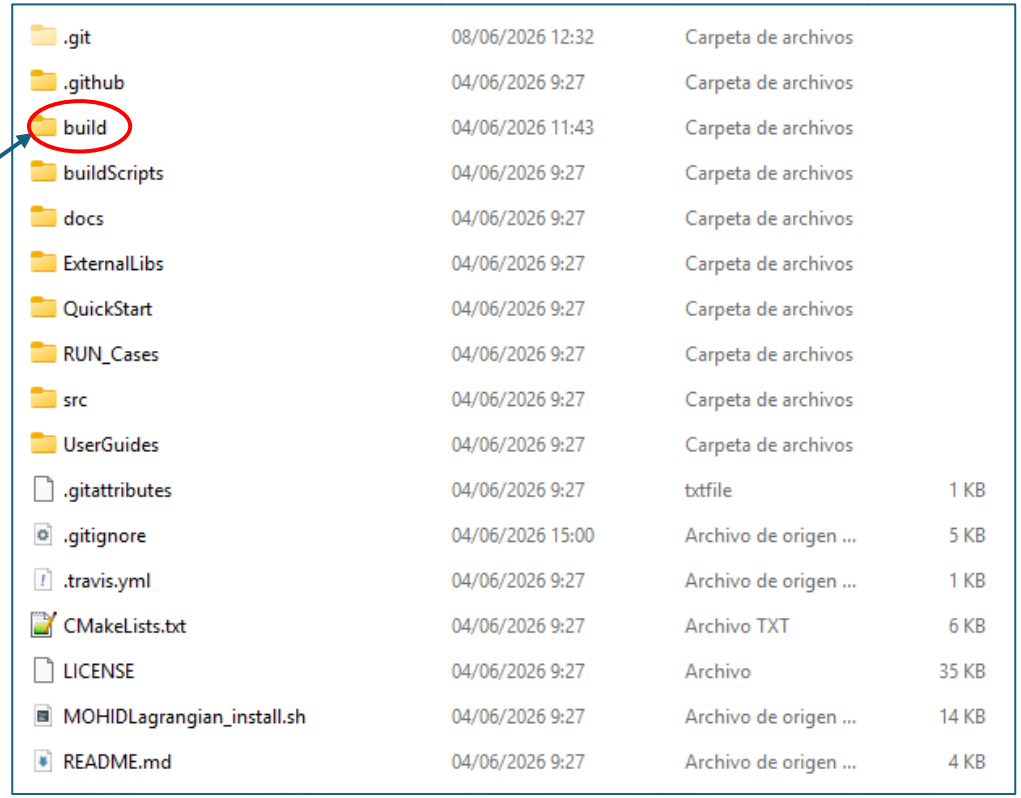
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Compiling MOHID Lagrangian:

- Open **buildScripts** folder and run `RunCmakeMOHIDLagrangian.bat`



.git	04/06/2026 9:27	Carpeta de archivos	
.github	04/06/2026 9:27	Carpeta de archivos	
buildScripts	04/06/2026 9:27	Carpeta de archivos	
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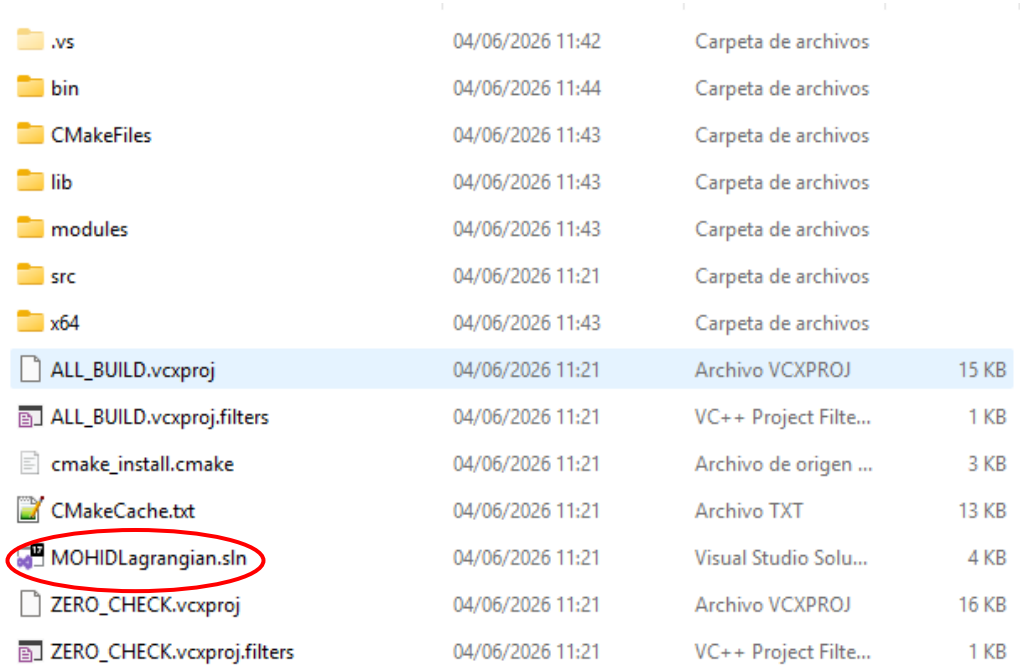


.git	08/06/2026 12:32	Carpeta de archivos	
.github	04/06/2026 9:27	Carpeta de archivos	
build	04/06/2026 11:43	Carpeta de archivos	
buildScripts	04/06/2026 9:27	Carpeta de archivos	
docs	04/06/2026 9:27	Carpeta de archivos	
ExternalLibs	04/06/2026 9:27	Carpeta de archivos	
QuickStart	04/06/2026 9:27	Carpeta de archivos	
RUN_Cases	04/06/2026 9:27	Carpeta de archivos	
src	04/06/2026 9:27	Carpeta de archivos	
UserGuides	04/06/2026 9:27	Carpeta de archivos	
.gitattributes	04/06/2026 9:27	txtfile	1 KB
.gitignore	04/06/2026 15:00	Archivo de origen ...	5 KB
.travis.yml	04/06/2026 9:27	Archivo de origen ...	1 KB
CMakeLists.txt	04/06/2026 9:27	Archivo TXT	6 KB
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MOHIDLagrangian_install.sh	04/06/2026 9:27	Archivo de origen ...	14 KB
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- Go into `build` folder

Compiling MOHID Lagrangian:

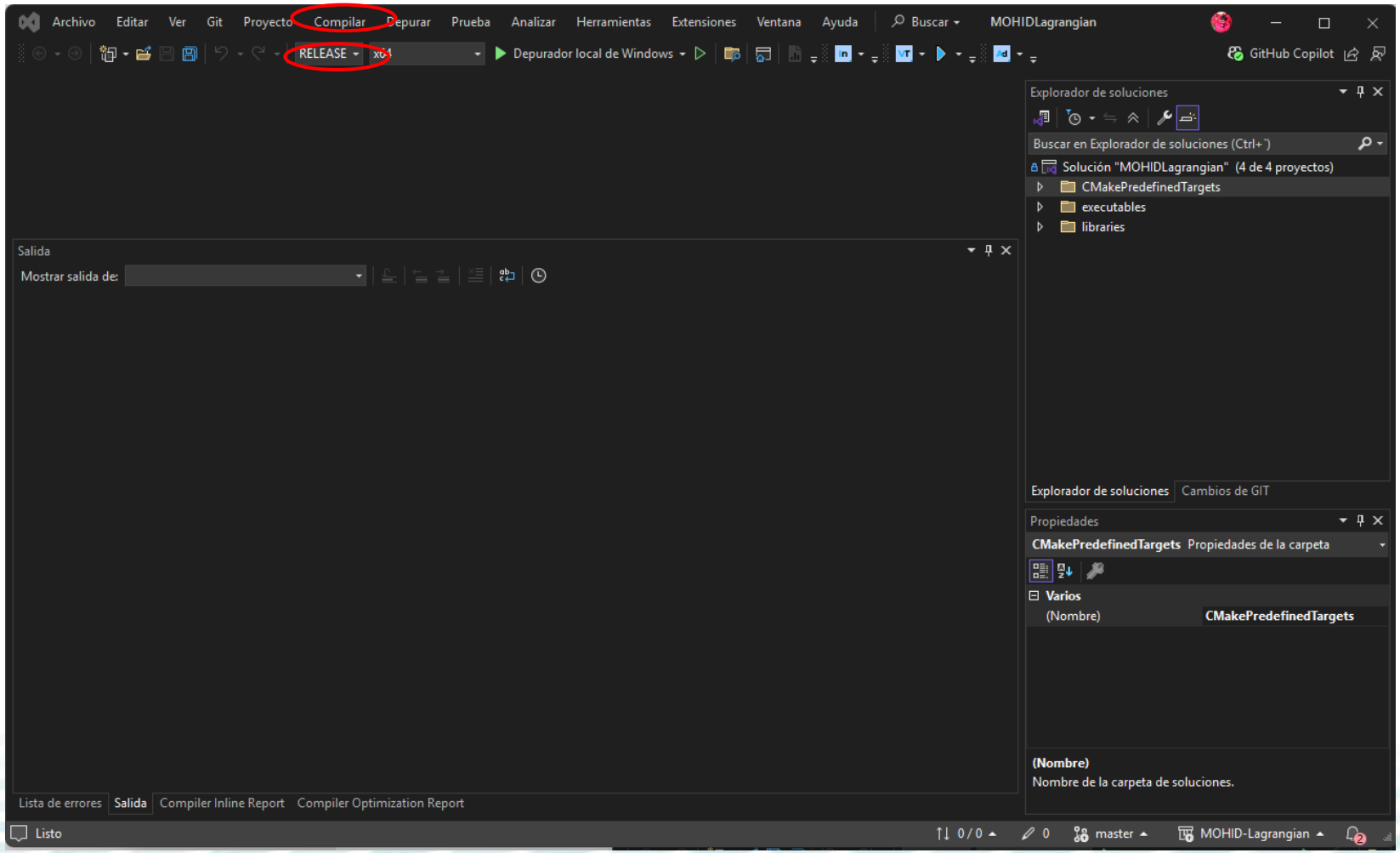
- Open `MOHIDLagrangian.sln` with Visual Studio 2020



.vs	04/06/2026 11:42	Carpeta de archivos	
bin	04/06/2026 11:44	Carpeta de archivos	
CMakeFiles	04/06/2026 11:43	Carpeta de archivos	
lib	04/06/2026 11:43	Carpeta de archivos	
modules	04/06/2026 11:43	Carpeta de archivos	
src	04/06/2026 11:21	Carpeta de archivos	
x64	04/06/2026 11:43	Carpeta de archivos	
ALL_BUILD.vcxproj	04/06/2026 11:21	Archivo VCXPROJ	15 KB
ALL_BUILD.vcxproj.filters	04/06/2026 11:21	VC++ Project Filte...	1 KB
cmake_install.cmake	04/06/2026 11:21	Archivo de origen ...	3 KB
CMakeCache.txt	04/06/2026 11:21	Archivo TXT	13 KB
MOHIDLagrangian.sln	04/06/2026 11:21	Visual Studio Solu...	4 KB
ZERO_CHECK.vcxproj	04/06/2026 11:21	Archivo VCXPROJ	16 KB
ZERO_CHECK.vcxproj.filters	04/06/2026 11:21	VC++ Project Filte...	1 KB

Compiling MOHID Lagrangian:

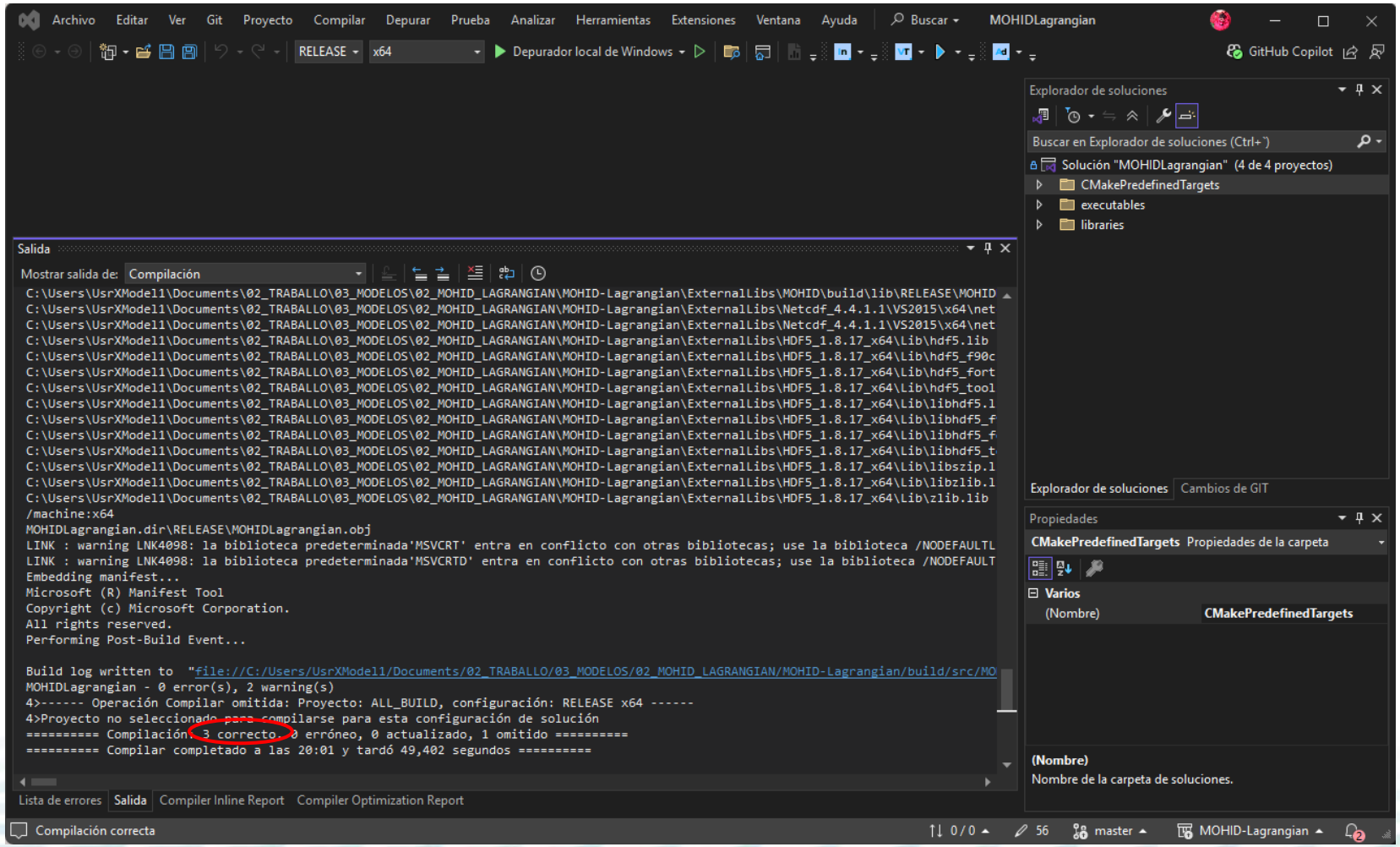
- Compile it with Visual Studio 2020 in RELEASE or DEBUG



Windows

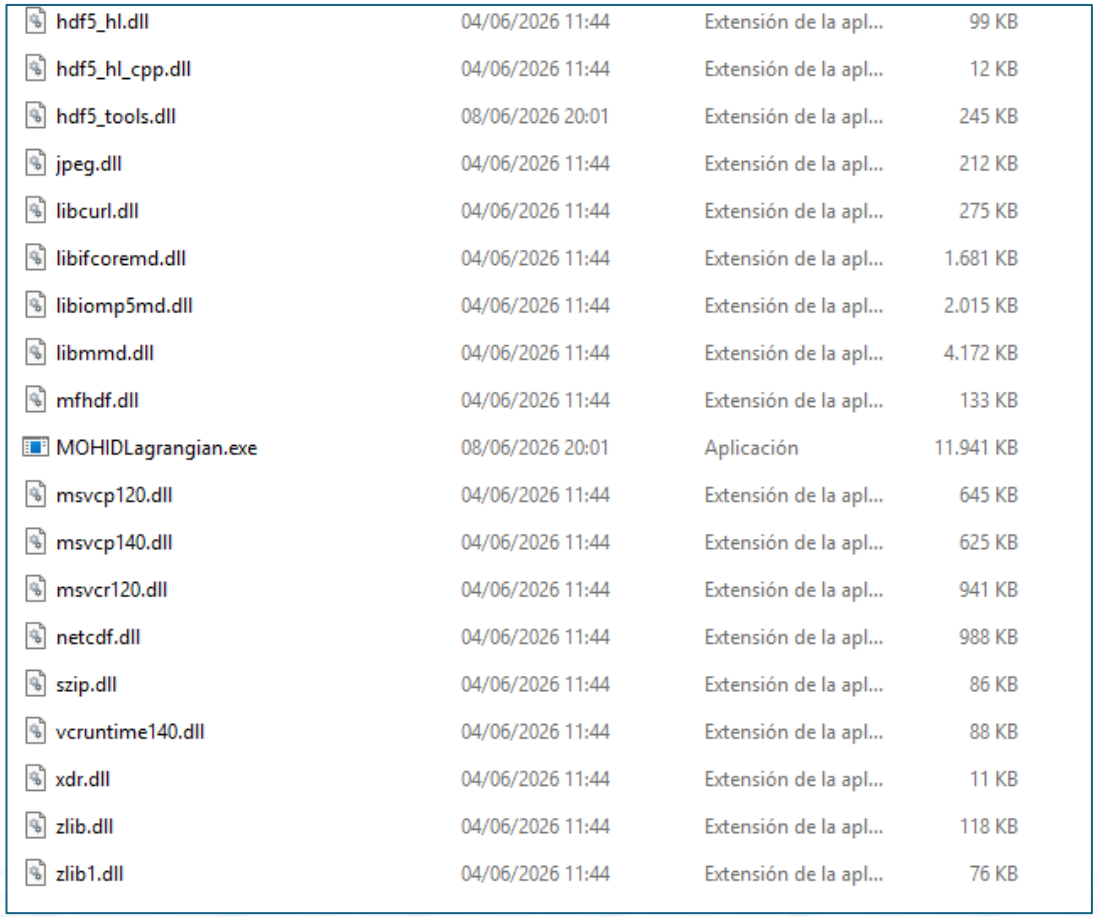
Compiling MOHID Lagrangian:

- Compile it with Visual Studio 2020 in RELEASE or DEBUG



Compiling MOHID Lagrangian:

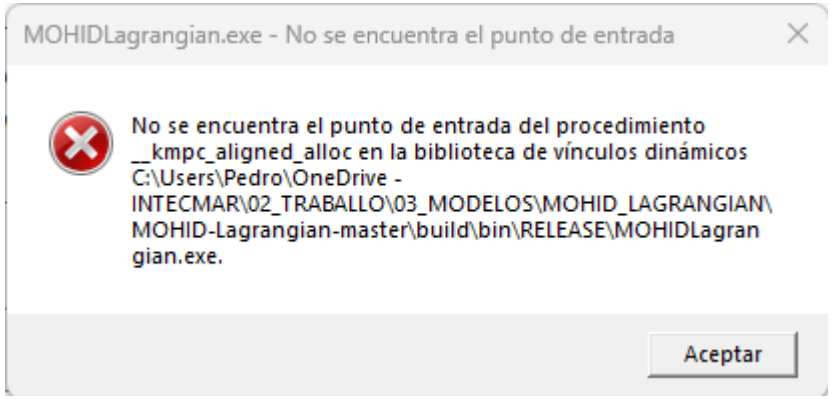
- Finally, the exe files are in build/bin/RELEASE or build/bin/DEBUG folders



hdf5_hl.dll	04/06/2026 11:44	Extensión de la apl...	99 KB
hdf5_hl_cpp.dll	04/06/2026 11:44	Extensión de la apl...	12 KB
hdf5_tools.dll	08/06/2026 20:01	Extensión de la apl...	245 KB
jpeg.dll	04/06/2026 11:44	Extensión de la apl...	212 KB
libcurl.dll	04/06/2026 11:44	Extensión de la apl...	275 KB
libifcoremd.dll	04/06/2026 11:44	Extensión de la apl...	1.681 KB
libiomp5md.dll	04/06/2026 11:44	Extensión de la apl...	2.015 KB
libmmd.dll	04/06/2026 11:44	Extensión de la apl...	4.172 KB
mfhdf.dll	04/06/2026 11:44	Extensión de la apl...	133 KB
MOHIDLagrangian.exe	08/06/2026 20:01	Aplicación	11.941 KB
msvcp120.dll	04/06/2026 11:44	Extensión de la apl...	645 KB
msvcp140.dll	04/06/2026 11:44	Extensión de la apl...	625 KB
msvcr120.dll	04/06/2026 11:44	Extensión de la apl...	941 KB
netcdf.dll	04/06/2026 11:44	Extensión de la apl...	988 KB
szip.dll	04/06/2026 11:44	Extensión de la apl...	86 KB
vcruntime140.dll	04/06/2026 11:44	Extensión de la apl...	88 KB
xdr.dll	04/06/2026 11:44	Extensión de la apl...	11 KB
zlib.dll	04/06/2026 11:44	Extensión de la apl...	118 KB
zlib1.dll	04/06/2026 11:44	Extensión de la apl...	76 KB

Compiling MOHID Lagrangian:

- The following runtime error may occasionally occur:



- To resolve this issue, copy `libiomp5md.dll` from `C:\Program Files (x86)\Intel\oneAPI\compiler\2024.2` *`(or similar)`* `\bin` to the executable's directory, overwriting the existing file.

Thank you!

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